STUDY OVERVIEW

# CacheWise replication and KV cache recovery

Serving coding agents under GPU memory pressure

01

## Replicate

Separate scheduling from eviction effects.

02

## Recover

Compare recomputation with host restoration.

03

## Schedule

Make request priority aware of host hits.

# The same prefix is useful again

**Model request**
Prefill + decode

→

**Tool runs**
GPU state competes for space

→

**Next request**
Old prefix + new context

## When the agent returns…

**Reusable prefix** | **+ new**

GPU eviction creates a recovery decision.

### Recompute

Run prefill again.

### Restore

Copy saved KV state back.

# Replication separates two decisions

## Scheduler × eviction policy

| | LRU | Predictive |
|---|---|---|
| **FCFS** | **FCFS / LRU** | **FCFS / predictive** |
| **Prefix-aware** | **Prefix / LRU** | **Prefix / predictive** |

## Coverage study

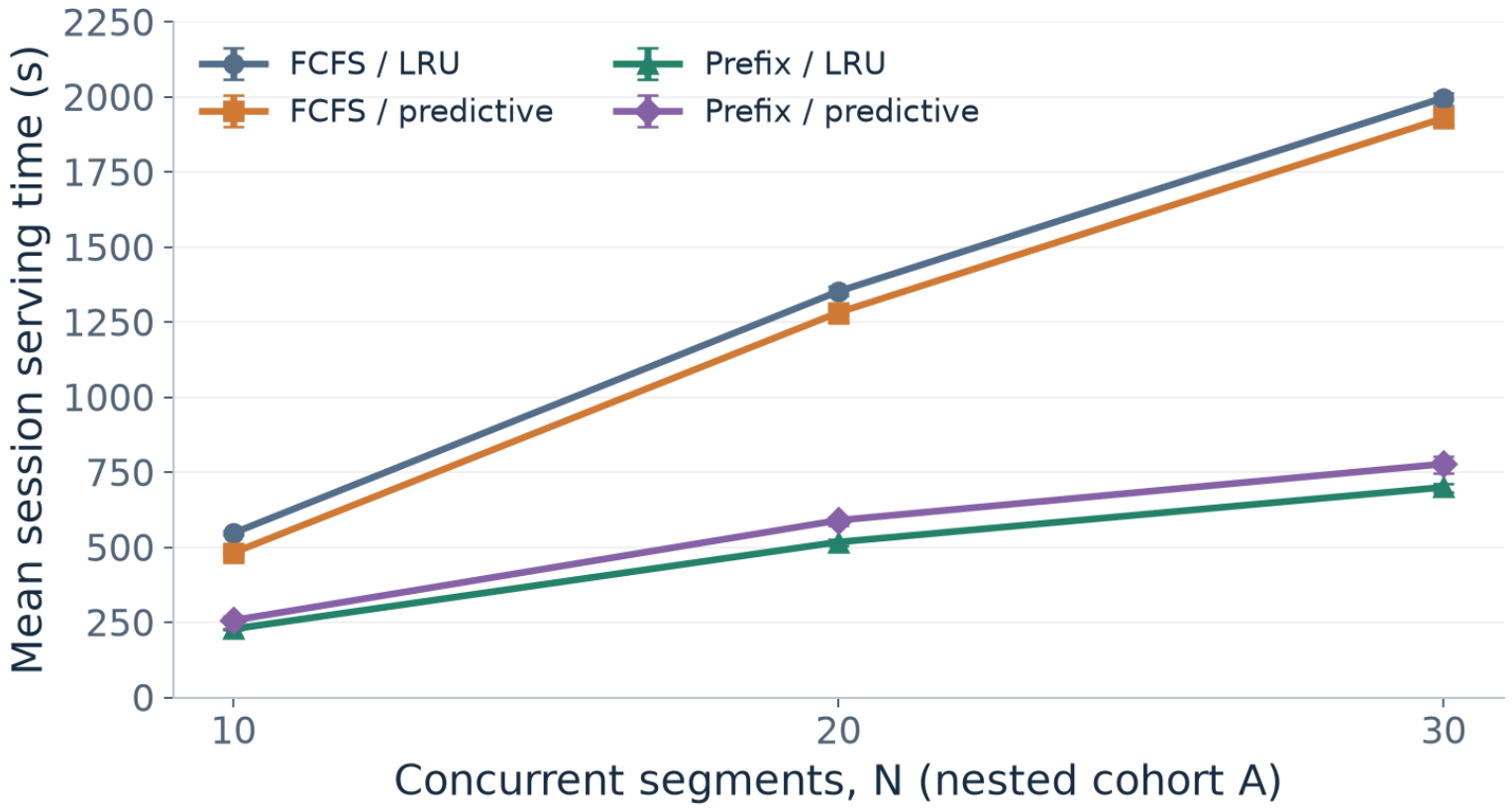
5 settings × 3 repeats
60 runs · 6,384 requests

14B model · BF16
287,696 KV token slots
512-token prefill chunks

**Primary metric: mean session serving time, excluding tool waits.**

Evidence: part1-coverage-comparison.json; report §3.1–3.2

# Prefix / LRU has the lowest mean



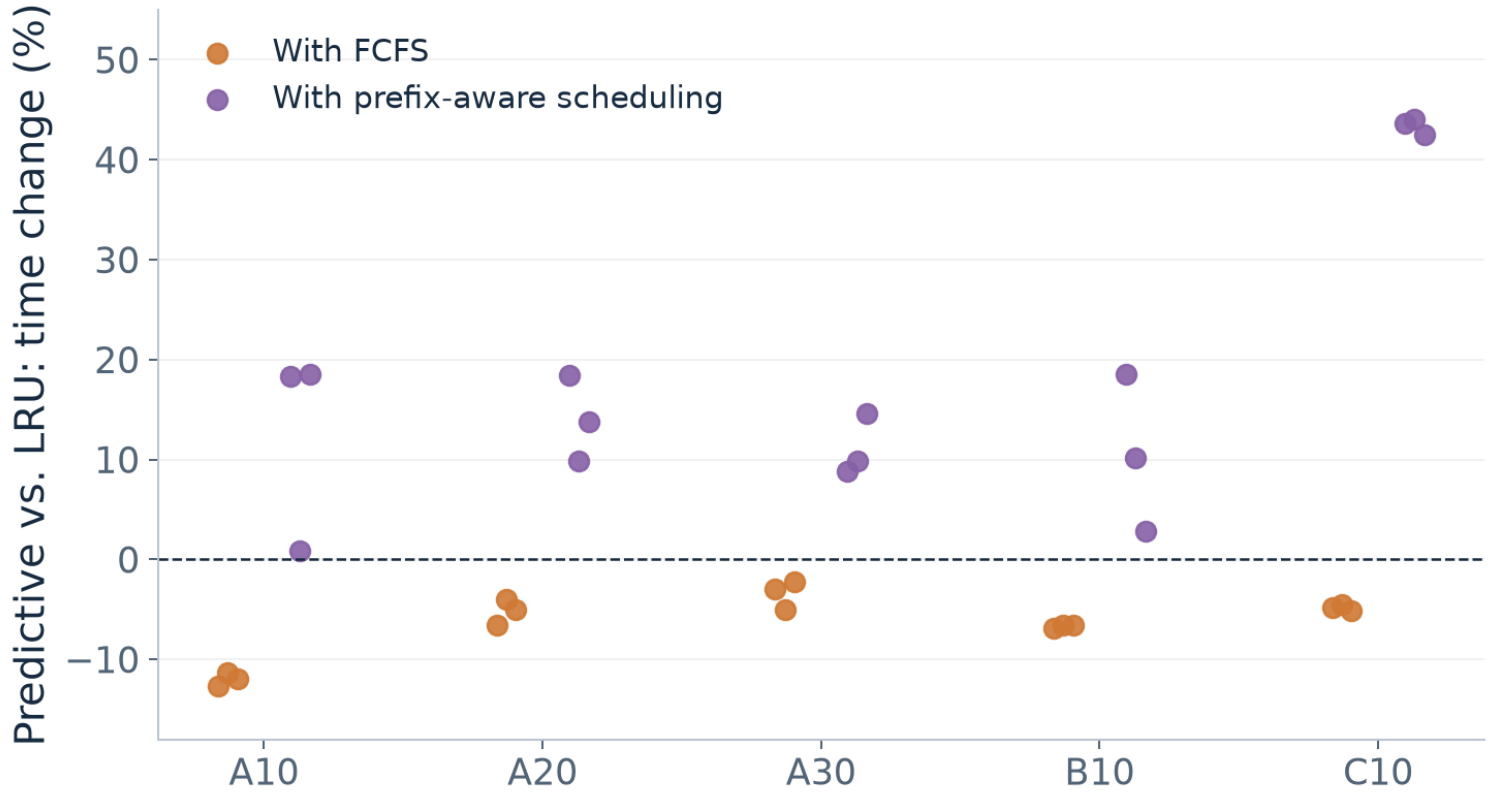

**5 / 5**

**settings favor prefix / LRU**

58–65% lower mean vs. FCFS / LRU in cohort A

Three repeats per cell. Ranges are observed minima/maxima; lower is better.

# Prediction depends on the scheduler




## Mean ≠ tail

A30 request P99
(average of 3 runs)

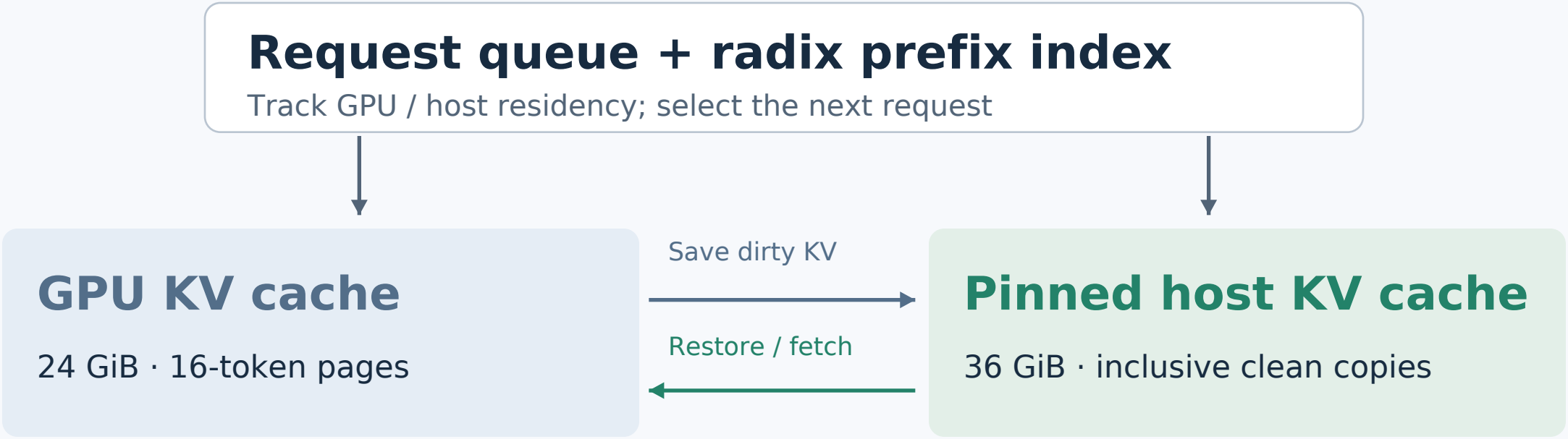
**713 s → 1,292 s**

FCFS / LRU →
prefix / LRU

Dots are matched-repeat effects. Positive values mean slower serving.

Evidence: matched coverage repeats and request tails; report Figure 1b and §3.3

# Preserve evicted KV in host DRAM




## On demand

Restore before prefill.

## Prefetch

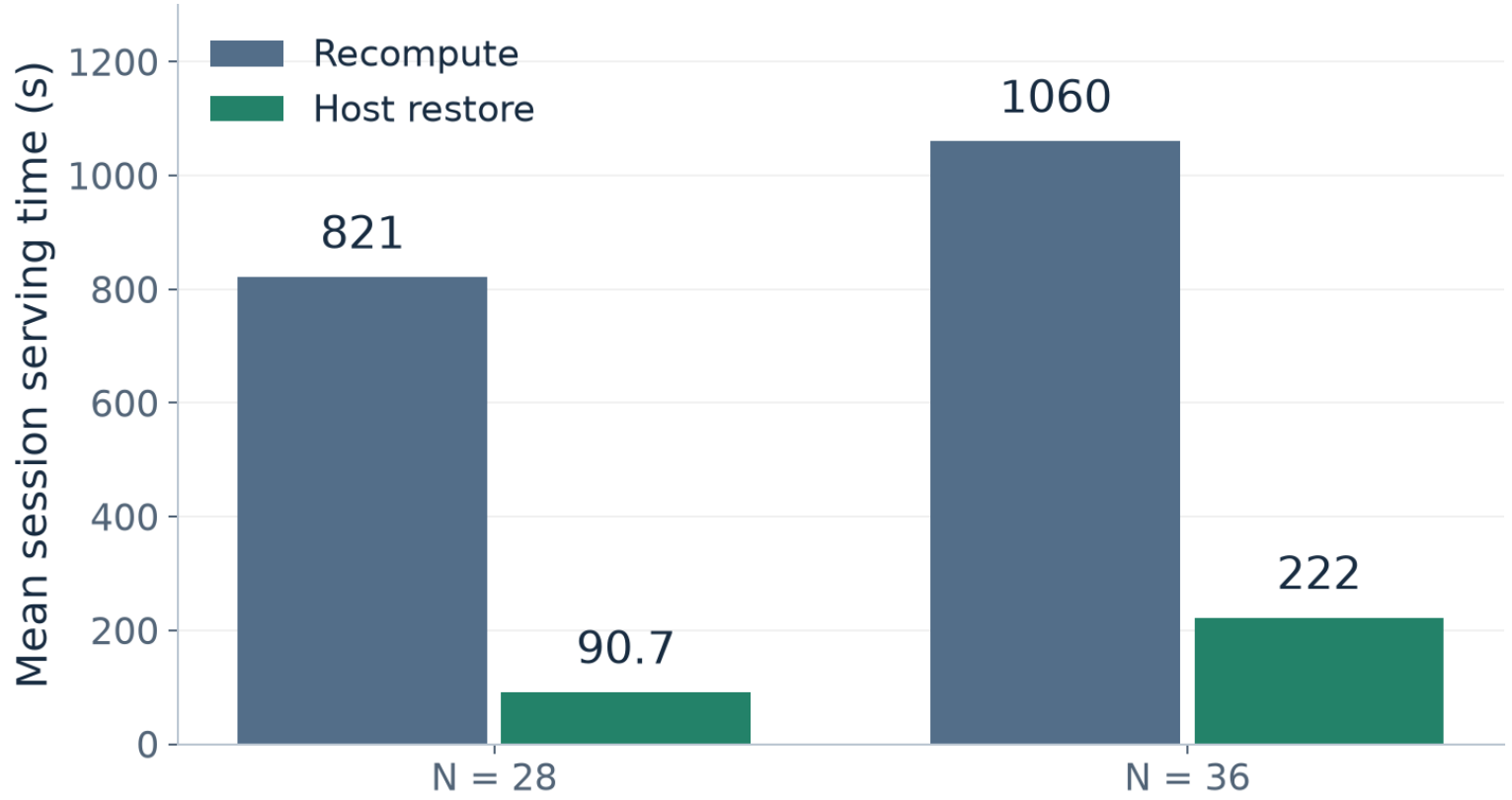
Overlap predicted reuse.

## Fallback

Recompute on a miss.

# Host restoration reduces session time

Recovery tradeoff: save + restore + management < recomputation




**9.1× / 4.8×**

reported time ratios
at N = 28 / 36

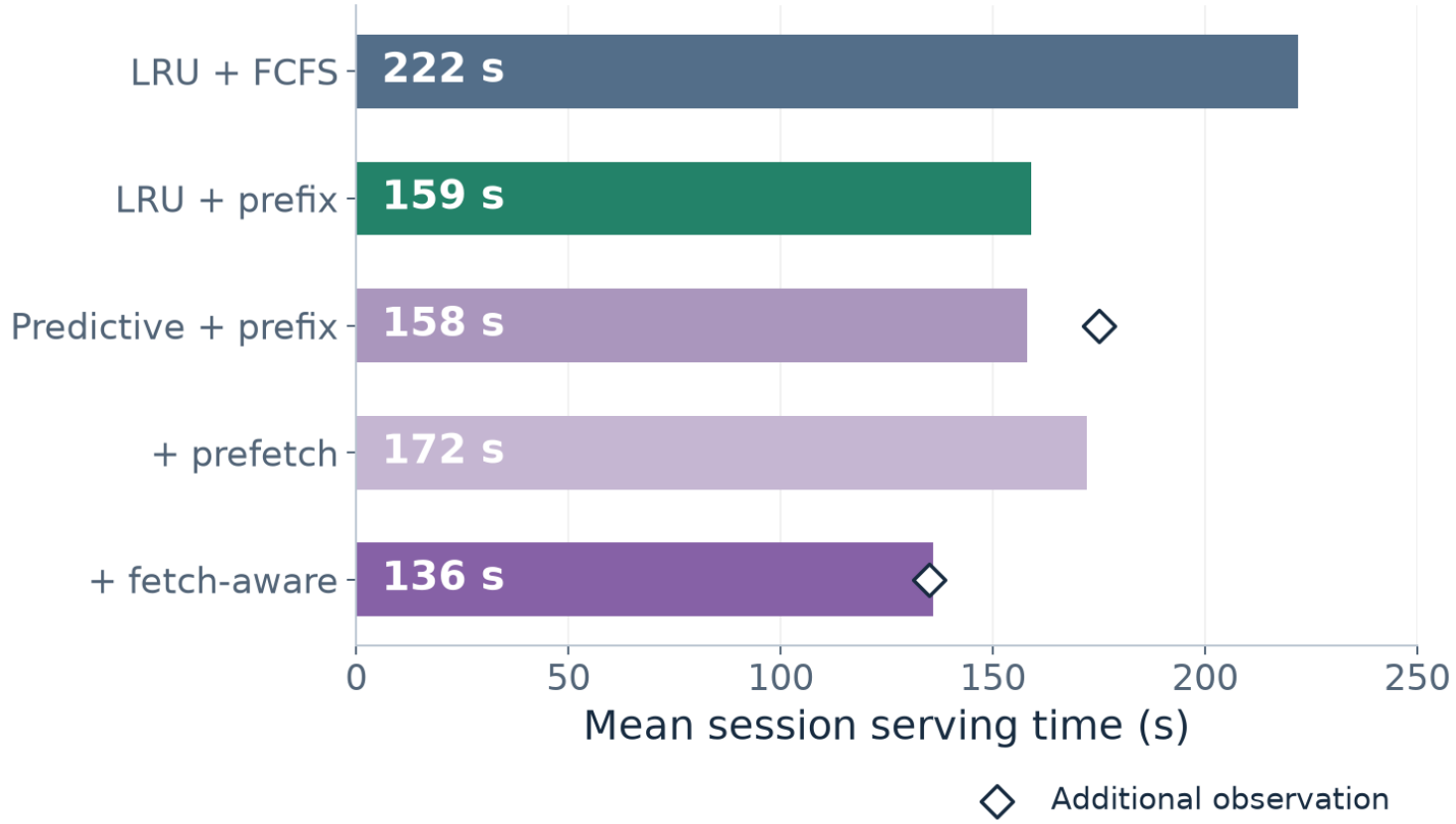
**+36 GiB host KV**
**24 GiB GPU KV**
**fixed**

One observation
per bar

Separate Mini-SGLang study. Added-capacity comparison; lower is better.

Evidence: host-dram-reported-results.json; report Figure 3a and §6

# Fetch-aware ordering discounts host hits




## Priority cost

**new prefill tokens + 0.05 × host tokens**

Mean of 2 observations
166.5 s → 135.5 s

Sparse repetitions; no confidence interval.

Bars: first observations. Diamonds: additional observations, not matched blocks.

Evidence: Mini-SGLangs 3f9aee7; report Figure 3b and Eq. (3)

# Keep the conclusions within the evidence

## Replication

Synthetic replay and selected segments.

One retained preemption; extra prefill work unmeasured.

## Host results

Separate engine and replay.

Added host capacity; most configurations run once.

## SSD extension

Not implemented or measured.

Needs full I/O, staging, metadata and recovery.

**Interpret both studies separately; retain the declared protocol deviation.**

# Three takeaways

## 01 Scheduling and eviction interact.

Prefix / LRU has the lowest mean in all five replication settings.

## 02 Host copies can avoid costly recovery.

Large reported gains include the added 36 GiB backing tier.

## 03 Order requests by recovery cost.

Fetch-aware results are promising; SSD evaluation remains future work.